

because of the good profitability itself, and the relative stability associated with it, but perhaps even more cogently because high earnings on capital generally go hand in hand with a good growth record and prospects. Thus what is really paid for nowadays in the case of highly profitable companies is not the good will in the old and restricted sense of an established name and a profitable business, but rather their assumed superior expectations of increased profits in the future.

This brings me to one or two additional mathematical aspects of the new attitude toward common-stock valuations, which I shall touch on merely in the form of brief suggestions. If, as many tests show, the earnings multiplier tends to increase with profitability—i.e., as the rate of return on book value increases—then the arithmetical consequence of this feature is that value tends to increase directly as the square of the earnings, but *inversely* the book value. Thus in an important and very real sense tangible assets have become a drag on average market value rather than a source thereof. Take a far from extreme illustration. If Company A earns \$4 a share on a \$20 book value, and Company B also \$4 a share on \$100 book value, Company A is almost certain to sell at a higher multiplier, and hence at higher price than Company B—say \$60 for Company A shares and \$35 for Company B shares. Thus it would not be inexact to declare that the \$80 per share of greater assets for Company B are responsible for the \$25 per share lower market price, since the earnings per share are assumed to be equal.

But more important than the foregoing is the general relationship between mathematics and the new approach to stock values. Given the three ingredients of (a) optimistic assumptions as to the rate of earnings growth, (b) a sufficiently long projection of this growth into the future, and (c) the miraculous workings of compound interest—lo! the security analyst is supplied with a new kind of philosopher's stone which can produce or justify any desired valuation for a really "good stock." I have commented in a recent article in the *Analysts' Journal* on the vogue of higher mathematics in bull markets, and quoted David Durand's exposition of the striking analogy between value calculations of growth stocks and the famous Petersburg Paradox, which has challenged and confused mathematicians for more than two hundred years. The point I want to make here is that there is a special paradox in the relationship between mathematics and investment attitudes on common stocks, which is this: Mathematics is ordinarily considered as producing precise and dependable results; but in the stock